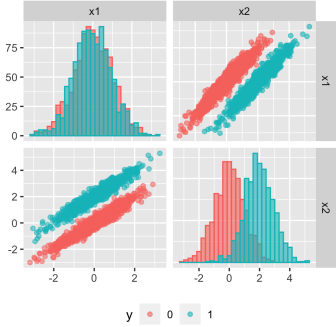
# Introduction to Machine Learning

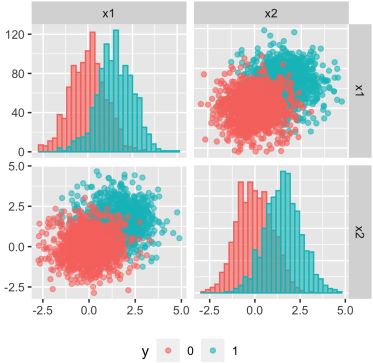
## Feature Selection
## Feature Selection: Filter Methods (Examples and Caveats)




**Learning goals**

- Understand how filter methods can be misleading
- Understand how filters can be applied and tuned

# FILTER METHODS CAN BE MISLEADING



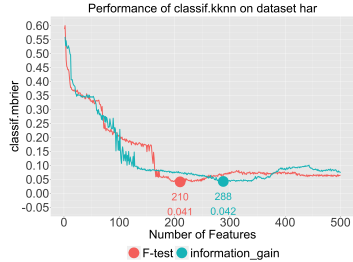
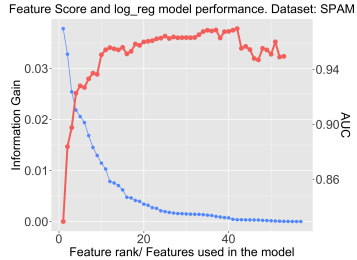

$\rho_{ACC}$ of log. reg. classifier with:

- feature $x_1$: 0.76
- feature $x_2$: 0.78
- both features: 0.85

**Information gain from presumably redundant variables**. 2 class problem with indep features. Each class has Gaussian distribution with no covariance. While filter methods suggest redundancy, combination of both vars yields improvement, showing indep vars are not truly redundant. For further details, see ▸ Guyon and Elisseeff 2003.

# USING FILTER METHODS

1. Calculate filter score for each feature $x_j$
2. Rank features according to score values
3. Choose $\tilde{p}$ best features
4. Train model on $\tilde{p}$ best features

**How to choose $\tilde{p}$?**

- Could be prescribed by application
- Eyeball estimation: read from filter plots
- Treat as hyperparameter and tune in a pipeline, based on resampling